Cogeneration Technology Adoption in the U.S.

by

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CES 16-30 **June, 2016**

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Abstract

Well over half of all electricity generated in recent years in Denmark is through cogeneration. In U.S., however, this number is only roughly eight percent. While both the federal and state governments provided regulatory incentives for more cogeneration adoption, the capacity added in the past five years have been the lowest since late 1970s. My goal is to first understand what are and their relative importance of the factors that drive cogeneration technology adoption, with an emphasis on estimating the elasticity of adoption with respect to relative energy input prices and regulatory factors. Very preliminary results show that with a 1 cent increase in purchased electricity price from 6 cents (roughly current average) to 7 cents per kwh, the likelihood of cogeneration technology adoption goes up by about 0.7-1 percent. Then I will try to address the general equilibrium effect of cogeneration adoption in the electricity generation sector as a whole and potentially estimate some key parameters that the social planner would need to determine the optimal cogeneration investment amount. Partial equilibrium setting does not consider the decrease in investment in the utilities sector when facing competition from the distributed electricity generators, and therefore ignore the effects from the change in equilibrium price of electricity. The competitive market equilibrium setting does not consider the externality in the reduction of CO2 emissions, and leads to socially sub-optimal investment in cogeneration. If we were to achieve the national goal to increase cogeneration capacity half of the current capacity by 2020, the US Department of Energy (DOE) estimated an annual reduction of 150 million metric tons of CO2 annually – equivalent to the emissions from over 25 million cars. This is about five times the annual carbon reduction from deregulation and consolidation in the US nuclear power industry (Davis, Wolfram 2012). Although the DOE estimates could be an overly optimistic estimate, it nonetheless suggests the large potential in the adoption of cogeneration technology.

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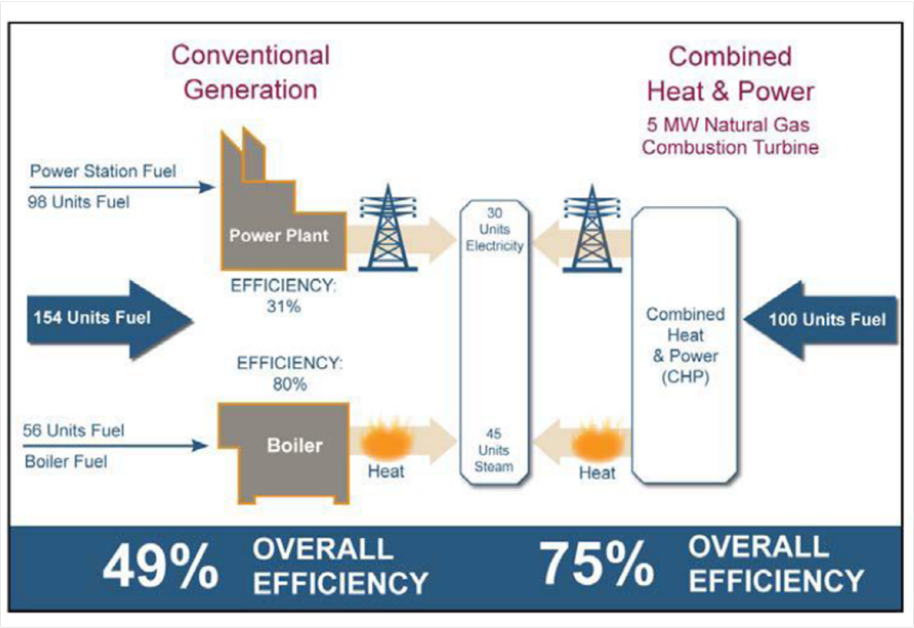
I am thankful for the advice and support from my dissertation committee: Steven Davis, Michael Greenstone, and Ali Hortascu. I am also thankful to Robert Burns, Steve Cicala, Thomas Covert, Cheryl Grim, Nathan Petek, Nolan Pope, Devin Pope, and seminar participants at the University of Chicago for comments that substantially improved the paper. Help from staff members at the US Census Bureau, US Energy Information Administration, is greatly appreciated. Support for this research at the US Census Research Data Center at Chicago from the University of Chicago is gratefully acknowledged. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. I am also grateful for the generous funding support from the Energy Policy Institute at University of Chicago.

1 Introduction

Cogeneration, also known as Combined Heat and Power (CHP), is a mature technology that produces electricity as a by-product when producing heat/steam. It essentially recycles the rejected/waste heat from the initial production of useful thermal output and uses it as input for electricity generation. This technology can be applied to industrial plants, commercial buildings, or even utilities, as long as the entity has need for both thermal and electric power. Conventionally, plants generate useful thermal output through a boiler, which loses much of its heat to the atmosphere, and purchase electricity from central power plants. With cogeneration, plants can make use of the waste heat to generate electricity on site, not only defray costs from electricity purchase, but also increase the overall efficiency and save on carbon emissions. U.S. Department of Energy (DOE) suggests that a CHP system can achieve 75% or above overall fuel efficiency (DOE, Figure 1). Central power plants typically operate in the 35% to 50% efficiency range. Depends on specific type of fuel used for steam and electricity generation, CHP system offer reduction in total primary fuel consumption by 25% to 35%. A natural gas fired CHP system can reduce CO_2 emissions close to 60% if it were to reduce electricity generation from a coal-fired power plant (Quinn et. al., 2013; DOE reports, 2012). These benefits of cogeneration have long been recognized by different countries across the globe. In Particularly, Denmark generated more than half of its electricity through cogeneration. Other Northern European countries, such as the Netherlands, Latvia, and Finland, all use cogeneration technology substantially to provide both electric and thermal power (IEA, Figure 2). In comparison, the U.S. only cogenerates about 8 percent of the total electricity in 2013, far below the average Northern European level, and far below its technical potential according to DOE's estimation.

Historically though, the US was the first to employ cogeneration technology and was a leader in its application. In 1900, over 59 percent of total US electricity generating capacity was at industrial sites, although not necessarily all through cogeneration. As utility-generated electricity became cheaper and more reliable, industry gradually shifted away from generating electric power on site. Meanwhile, various federal and state statutes were passed in the early 1930s to encourage vertically integrated structure in the utility sector and discourage decentralized electricity generation. By 1950, onsite industrial generation declined to about 17 percent of total capacity. This number was about 3 to 4 percent by 1980. Cogeneration potential (industry thermal load fit for cogeneration), during the same period, increased at much faster pace than the adoption rate. Three major institutional obstacles made it difficult for cogenerators to make their investment economically profitable. First, utilities were reluctant to purchase the electricity from the cogenerators they were interconnected to. Second, utilities charged high standby rates. And third, cogenerators that sold electricity risked being classified, and therefore under state and federal regulation, as electric utility (Ch. 7,

Figure 1: Fuel Savings From a CHP System

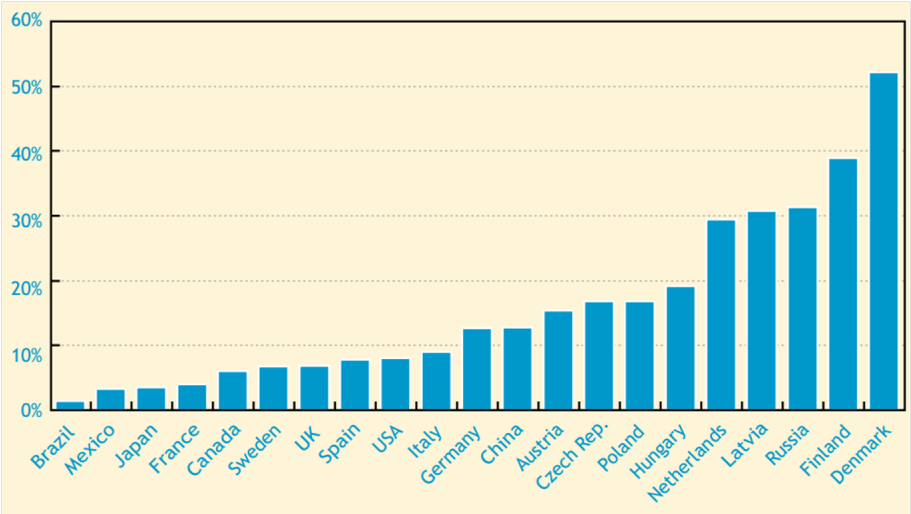


Source: U.S. Department of Energy, and U.S. Environmental Protection Agency, 2012

Industrial and Commercial Cogeneration).

Recognizing the benefits of cogenerators and other fuel efficient generators, as well as being part of national security efforts to become less dependent on imported fuels, the passing of the 1978 Public Utility Regulatory Policies Act (PURPA) turned the situation more favorable for cogenerators by removing all three of the institutional obstacles. Once a cogeneration facility or a small power production facility is certified as a qualifying facility (QF) under PURPA, it enjoys several statutory benefits. Three main benefits are as follows: first, utilities are mandated to interconnect with QFs; second, host utilities are required to purchase electricity ("must purchase" obligation) generated from QFs at utility's "avoided cost" – a unit price that is at least as much as the unit generating cost at the utility; and third, host utilities have to provide electric service ("must sell" obligation) indiscriminately to any QF at its requested capacity. "Avoided cost" is formally defined as "the incremental cost to the electric utility of electric energy or capacity or both which, BUT FOR the purchase from the QF or QFs, such utility would generate itself or purchase from another source". State and non-regulated utilities always determine avoided cost, either by administratively determining them or through market-based methods. "Must purchase" obligation applies to all electric utilities, and each federal power marketing authority, unless the Federal Energy Regulatory Commission (FERC) grants a wavier (Burns and Rose, 2014). In essence, "must purchase" obligation is a put option and "must sell" obligation is a buy option provided to the qualifying cogenerators, that they can choose to sell/purchase electricity at higher/lower prices to/from willing buyer/seller other than their host utility, but they always have the option

Figure 2: CHP share of national power production



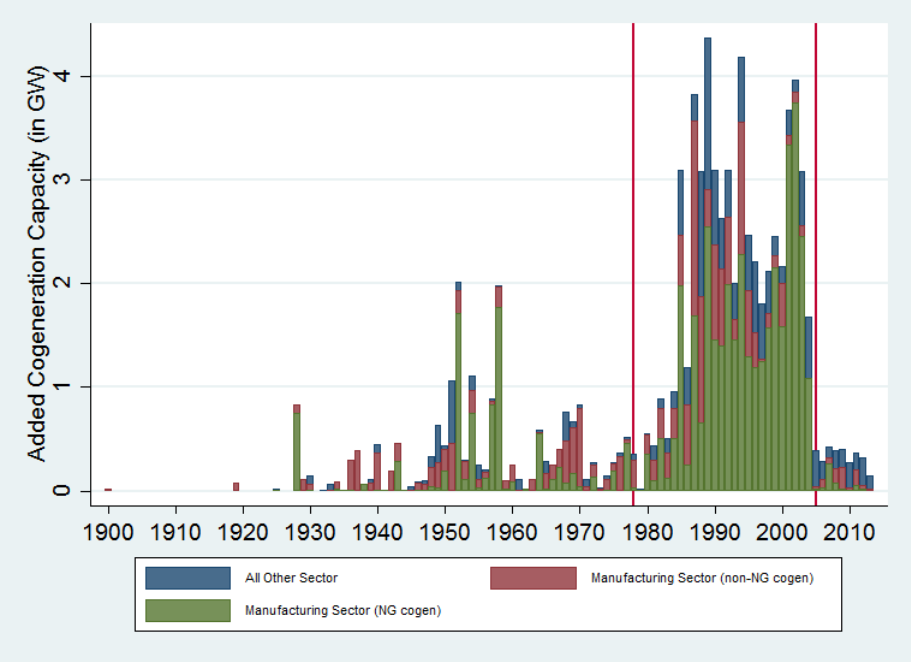
Source: International Energy Agency (IEA), 2008

to sell/buy at guaranteed prices from their host utility. The thought at that time was that this put option at "avoided cost" promoted the development of QF power without incurring uneconomical subsidies. Whether or not this "avoided cost" created arbitrage opportunities can be debated, but both side agree that the new statutory environment created by PURPA encouraged the adoption of cogeneration technology. Indeed, we observe a sharp increase in the installation of cogeneration soon after 1978 in the data (ICF International, Figure 3).

However, this increase in cogeneration capacity was somewhat stalled after 2005. Out of the current 82 GW cogeneration capacity, less than 4 GW is added from 2005 to 2013, compared with the 4.8 GW capacity added in just two years from 2003 to 2004. This makes achieving the national goal of 122 GW cogeneration capacity by 2020 seem dim. But if the goal were met, the U.S. Department of Energy and Environment Protection Agency estimated that it would save energy users $10 billion a year compared to current energy use; save one quadrillion Btus (quad) of energy – the equivalent of 1 percent of all energy use in the U.S.; reduce emissions by 150 million metric tons of CO_2 annually – equivalent to the emissions from over 25 million cars; and result in $40-$80 billion in new capital investment in manufacturing and other U.S. facilities over the next decade. Understanding the factors behind cogeneration technology adoption will not only help to understand the drop in adoption after 2005, but also help policy makers to effectively promote the technology and reach their ambitious goal.

Preliminary results suggest that relative input prices have limited importance in shaping adoption decisions among manufacturing plants. The regulatory environment might have been more important. In 2005, under intense lobbying by electric utilities, the Energy Policy Act 2005 (EPAct 2005) raised the fuel

Figure 3: CHP Installation 1900-2013



Source: Data from ICF International Combined Heat and Power Installation Database. Each bar represents the cogeneration capacity added in a given year.

efficiency requirements for cogeneration QF status and relaxed the "must purchase", "must sell" obligations for host utilities. Specifically, no utility must-purchase obligation exists if FERC finds that the QF has nondiscriminatory access to wholesale electricity markets and any QF larger than 20 MW is rebuttably presumed to satisfy the requirement. Conveniently, although these obligations might not exist any more, the utility buy-back and standby rates were not altered by the EPAct 2005. A regression discontinuity design would be ideal for analyzing the economic effect of "must purchase" obligation on both the extensive margin of cogeneration technology adoption and intensive margin of cogeneration technology utilization. Similarly, no utility must-sell obligation exists if FERC finds that competing retail electric suppliers are willing and able to sell and deliver electric energy to the QF, but there is no rebuttable presumptions on plant size. In general, host utilities complain more about the must-purchase obligation than the must-sell obligation, because unlike "avoided cost" (buy-back rates) set by state and non-utilities, standby rates are always set by individual utilities. The interconnection rules stayed relatively unchanged prior and after EPAct 2005. Thus, among the bundled benefits for being a QF, the change in "must purchase" obligation likely affected the cogeneration plants the most.

Preliminary results also show that plants do change their behavior significantly after the adoption of cogeneration. They purchase less electricity and consume more natural gas (primary fuel for generating thermal output), which is expected. Holding the total amount of electricity consumption fixed, the more we

generate on site, the less we need to purchase. Over 70 percent of all cogeneration plants are fired by natural gas. Majority of the adopters already use natural gas in some extent prior adoption. Recall that cogeneration system requires less fuel to generate both thermal and electricity output than if we were to generate thermal and electricity separately from traditional boiler and power plants. But for the same amount of thermal output, cogeneration systems would require more natural gas than traditional boilers. Hence, to maintain process steam production, we would expect the adopting plant to raise their consumption of natural gas. More cogenerated electricity there is in a 3-digit zip code area, the lower is the average electricity prices. The effect of cogeneration on the investment, fuel consumption, and carbon emissions in the utilities sector still need to be studied, in combination with the within-plant adoption effects.

The research design is to first estimate the adoption elasticities with respect to energy prices and regulatory factors, also estimate the crowding out effects of electricity generation (as well as capital investment) in the utilities sector, and the size of the environmental externalities associated with cogeneration. Then given these parameters we estimated using micro level data, we can attempt to do some calibration and simulate several scenarios with general equilibrium effects in place. This can help provide guidance on what would happen to the equilibrium electricity price, fuel consumption, and carbon emissions if we were to increase the cogeneration electricity share to different desired levels. Section 2 will briefly offer a simple model of cogeneration. Section 3 will provide information on available data resources. Section 4 will introduce my empirical strategies and discuss the preliminary results. Section 5 concludes and details future research design.

2 A Simple Model of Cogeneration

From the engineering perspective, cogeneration system is most efficient when the power and steam demand are balanced at all times (GAO, 1980). In practice, this is difficult to do and most plants choose their cogeneration capacity primarily based on their steam load. Individual cogeneration plants set up arrangements with local utilities to purchase electricity when the steam load is low and does not generate sufficient electricity for plant operation, and to sell excess cogenerated electricity when the steam load is high. So meeting the steam demand is the dominant consideration for non-utility adopters of cogeneration technology, while efficient power generation is the dominant consideration for utility. Therefore, for electricity generation alone, utilities are more efficient using the same amount of fuel as cogenerators.

The primary function for switching from a conventional boiler to a cogeneration system is to recycle rejected/wasted heat from steam (or other useful thermal output) generation. There is no theoretical reason to believe this technology switch would affect the actual amount of electricity usage, or final output (total

value of shipment), or plant productivity. Preliminary results also confirm this understanding. If we think of convert fuel input to final goods in two steps, first to convert fuel to steam and electricity, then use steam, electricity, and other intermediate inputs to generate the final output, both the results and intuition suggest that the adoption of cogeneration change only the first step in the production process and not the second. Holding final output and plant productivity fixed, the input requirements should be the same prior and after cogeneration technology adoption.

This allows us to model the adoption decision at individual plants abstracting from the second stage in the production process. Consider the following discrete choice model with fixed input requirements[1]. In order to generate the same amount of steam and electricity, a plant with conventional technology o has fixed input requirements of E_o units of purchased electricity and G_o units of natural gas (or other type of fuel it uses to generate steam), whereas a plant employing cogeneration technology c has fixed input requirements of E_c units of purchased electricity and G_c units of natural gas. $E_o > E_c$ and $G_o < G_c$ since cogeneration plants need more natural gas as input to generate both steam and electricity, thus the expected quantity of electricity purchased would be lower.

The per-period variable profit function for a type α plant ($\alpha \in \{o, c\}$) is

$$\pi_\alpha(P, m) \quad = \quad P - (P_E^P E_\alpha + P_G G_\alpha) + P_E^S f_\alpha(G_\alpha) \tag{1}$$

where P is the final good price that is exogenous to the plant since plants are price takers; m is market condition (a price vector that includes all input prices); P_E^P is the purchase price of electricity; P_E^S is the sell price of electricity; P_G is the purchase price of natural gas; function $f_\alpha(G_\alpha)$ is the amount sold out of the cogenerated electricity. $f_o(\cdot) = 0$ assuming without cogeneration, no electricity is generated. Quantity of final output is omitted since the adoption decision is independent of the second stage production for any individual plant in the simple model. In reality, however, the scale effect matters. When steam load factor is too low, the capital related cost will outweigh fuel related cost, which makes the system uneconomical to operate. The sets of plants suitable for this model are those with enough steam load requirements and have the technical potential to be an adopter of cogeneration technology. In my empirical analysis, I control for the plant hours by quintile indicators[2], allowing the elasticity parameters be non linear in plant scale.

[1] Model assumes cogenerated electricity is tied proportionally to plant's steam load, and does not allow plants to choose the amount of electricity to be cogenerated on site (unlike Rose et. al 1990, 1991). This is reasonable since the my primary objective is to understand the adoption decision making process, not the utilization of the technology. Also, from the engineering side, the proportionality assumption is reasonable as well since that is the most efficient use of the technology.

[2] Although the manufacturing data at US Census Bureau do not have steam load figures, plant hours capture load demand. When load factor is high, that means the plant runs close to 8760 hours a year, as maintenance requirements permit (GAO, 1980).

Any incumbent plant making a decision to switch technology from conventional to cogeneration would have to consider the fixed cost of installing the technology. Suppose market condition goes from m to $\bar{m}$ and persists forever, with discount rate β, fixed cost F, a plant would switch to cogeneration if and only if

$$\frac{\pi_c(P,\bar{m})}{1-\beta} - F \quad > \quad \frac{\pi_o(P,\bar{m})}{1-\beta}$$

This result is intuitive. It says an incumbent plant with conventional technology would only switch to cogeneration if the cumulative net profit generated by cogeneration discounted to its present value is more than the fixed cost of installation. Notice that this is a technology switching problem, the fixed cost of installing the conventional technology in the first place is already sunk at this point. Insert per-period profit function (eqn 1) into the above condition, the net profit generated by cogeneration can be break down into three components: purchased electricity cost saved, plus additional revenue generated by selling electricity, then subtracts off the additional cost on natural gas consumption.

$$\underbrace{P_E^P(E_o - E_c)}_{electricity\,cost\,defrayed} + \underbrace{P_E^S f_c(G_c)}_{additional\,revenue\,generated} - \underbrace{P_G(G_c - G_o)}_{additional\,cost\,on\,natural\,gas} > \underbrace{(1-\beta)F}_{discounted\,fixed\,cost} \tag{2}$$

As P_E^P goes up, all things equal, we would expect an increase in the adoption rate. Similarly, a plant that expects to generate more electricity on site is more likely to adopt the technology. $E_o - E_c$ kind of captures the steam load scale effect. With a high steam load, plants with same E_o can expect a large difference in $E_o - E_c$ assuming variety of cogeneration technologies are equal efficient in converting wasted heat into electricity. My empirical design in Section 4 heavily depends on this simple model in the background.

Thus far, the model only speaks to decentralized adoption (technology switching) decision making for incumbent non-utility plants. Its goal is to provide clear theoretical guidance to what should be included in the regression analysis, instead of a structural estimation model.

3 Data

This primary data source for this project is from the US Census Bureau[3] and the US Energy Information Administration. The US Census Bureau has detailed information on manufacturing plants that has been exploited greatly by researchers. However, the energy usage dimension of the data has not been touched

[3]Author is under Special Sworn Status to use the confidential data from the US Census Bureau.

Table 1: Cogeneration Capacity Added

Time Period	(1)	(2)	(3)	(4)
1900-1978	18.91	23.03%	16.17	85.49%
1978-2005	60.32	73.46%	48.81	80.94%
2005-2013	2.88	3.51%	1.29	44.64%
Total	82.11	100%	66.27	80.71%

Notes: Column 1 lists the cogeneration capacity (GW) added. Column 2 lists the share of cogeneration capacity added in terms of the total cogeneration capacity installed as of 2013–82 GW. Column 3 lists the cogeneration capacity (GW) added in the manufacturing sector. Column 4 lists the share of manufacturing cogeneration capacity added in terms of the same-period overall cogeneration capacity added. Data is from the ICF International Combined Heat and Power Installation Database; last updated in July, 2013. Maintenance of this database is supported by the U.S. Department of Energy and Oak Ridge National Laboratory.

nearly as much. In combination with labor, capital, and productivity data at the plant level, we can paint a broader picture of plant performance when studying the effect of a change in energy efficiency. More importantly, majority of the cogeneration technology employment is in the manufacturing sector. Table 1 column 3 shows that overall, roughly 80 percent of the economy wide cogeneration capacity in the US is in the manufacturing sector. From 1985 to present (most recent 2010), in three- to four-year intervals, the Census Bureau distributed and collected data from the Manufacturing Energy Consumption Survey (MECS), which not only ask about annual fuel consumption, but also technology on site, specifically the cogeneration technology. Supplementing the MECS with Annual Survey of Manufactures (ASM), the Census of Manufactures (CM), and the Longitudinal Business Database (LBD), we have information on plant's 6-digit NAICS code, age, size (by value of shipment or employment size), plant hours, zip code, county, exit year, annual expenditure on various fuel use, quantity of fuel purchased, presence of cogeneration technology, and amount of electricity cogenerated on site. Since MECS only have a small certainty sample, we do not have a large balanced panel. But for each cross section, we have roughly 6,000-10,000 plants. In total we have roughly 72,000 plant-year observations. However, a large portion of these plants only appeared once in the data set. Average number of appearance in the data is lower than three. Most of these plants are in energy intensive industries and bigger in size than the average manufacture plant in the universe of manufactures (CM). The estimation of manufacturing plant's adoption elasticity with respect to energy input prices uses entirely the data from within the Census Bureau. However, there is no data on the utilities sector, which makes estimating the effect of industrial cogeneration on utilities impossible.

The US Energy Information Administration (EIA) fortunately has detailed information on all electricity generators above 1 MW or greater of combined nameplate capacity, which means both non-utilities and utilities are included in the sample. [4]For utilities, the EIA has annual data from 1970s to present, with

[4]Some cogenerators have smaller capacity than 1 MW. But in total, missing these plants do not make up a large difference. The ICF International Combined Heat and Power Installation Database have very basic capacity and location information for each cogenerator regardless of size. US Department of Energy uses this data to produce the 82 GW current capacity figure in 2013. Data from EIA disagrees with the 82 GW figure, and only have roughly 70 GW. From author's calculation using ICF database, total cogenerator capacity from plants with less than 1 MW generating capacity amounts to less than 0.6 GW. It

information on name plate capacity, first year in operation, fuel expenditure, quantity purchased, and net electricity generated. For non-utilities, the EIA started to collect the data quite a bit later, almost 20 years later. But with information on first year in operation, we can back track some years ago. The data also indicates whether the non-utility uses cogeneration, and whether it is FERC certified cogeneration QF, which the Census data does not have. This is particularly useful for analyzing the policy effect on cogeneration utilization and adoption. Due to confidentiality protection, the EIA does not publish fuel cost data for non-utilities. Other variables that are collected in the survey forms but not publicly available are revenue and equipment costs. These information would undoubtedly increase our understanding of cogeneration technology in the US. But for now, I hope to use the already available data at hand to establish and demonstrate the impact of non-utilities cogeneration adoption on utilities and the environment. If the results deem promising, then it might be worthwhile to go through the confidential data access application process, which may take more than a few months.

The EIA non-utilities data include not only the cogenerators in the manufacturing sector, but also in other sectors such as schools and commercial buildings, which represents over half of the technical potential for future cogeneration adoption according to US Department of Energy estimates[5]. Thus, the EIA data is more comprehensive in the coverage of cogeneration plants, while the Census data is better positioned in studying the technology in manufacturing sector. In combination of both, we can hope to provide a complete picture about the development and employment of cogeneration for the past 40 years, and be able to address the proposed research questions with sufficient data[6].

4 Empirical Strategies and Preliminary Results

At the current moment, the EIA data still needs some preparation to be ready for analysis. The MECS carries the backbone for most of the work presented in this paper. Although MECS was originated in 1985, it is not until 1994 the survey started asking question on the cogeneration technology. But the survey did consistently ask about the amount of electricity cogenerated on site throughout 1985 to present. Hence, four definitions of technology adoption are used in the analysis: usage adoption, direct adoption, narrow

remains unclear where the differences might be coming from. But since EIA is the main agency in charge of collecting electricity generator level data, and plants with less than 1 MW capacity seems to make up a small portion of the total capacity, I will use the EIA data for empirical analysis.

[5]See Figure 9 in Appendix

[6]Technically, one can link the EIA data with Census data at plant level. Since the two institutions use different plant identifier, merging would need to be done using name and location of the plants. I do not have access to the names of the plants at Census, and would need additional application and data clearance. Therefore, for now, when using data from both sources, I aggregate plants up to 3 digit zip code level. Ideally, with the confidential data from both agencies, we would be able to construct a data set that covers from 1970s to present on annual basis (instead of MECS every three- to four- year, and from 1985 onward) of detailed energy consumption and technology usage among manufacturing plants as well as power plants in the utility sector. This would greatly increase the size of current unbalanced panel used in the analysis both cross-sectionally and longitudinally.

adoption, and broad adoption. Usage adoption is defined as a plant going from cogenerating zero electricity to positive amount (data from 1985-2010). Direct adoption is defined as a plant going from reporting not having any cogeneration technology present to having at least one type of cogeneration technology present (data from 1994-2010). Narrow adoption takes the intersection of the two and braod adoption takes the union of the two. For all these adoption definitions, the adoption indicator turns on if and only if the plant reports a non user for both the previous survey year and the year prior to that, and a user of the technology for two consecutive periods after. A few plants adopted more than once by broad and usage adoption, but very few adopted more than once by narrow or direct adoption definitions. Given the nature of cogeneration, if a plant operates at all, it will have to generate both steam and electricity, unless the plant has a separate conventional boiler. But since cogeneration is much more efficient than the conventional boiler without much increase in machine start up cost, it is hard to imagine a plant would not use the cogeneration to generate their steam load. It is puzzling to have plants report having the technology on site but cogenerated zero electricity, if it were not reporting error. This problem could potentially be resolved if we merge the EIA data at plant level since they specialize in electricity generation side of the business reporting. For now, I use all four definitions and hope to obtain a consistent picture across definitions. It turns out that narrow and direct adoption produce close to identical results, while broad and usage adoption produce similar results.

The two primary objectives I aim to achieve using the Census data is to estimate the adoption elasticities with respect to energy price shocks, and to estimate the fuel consumption elasticities with respect to cogeneration adoption, all in the manufacturing sector. The EIA data will help us understand the effect of adoption on utility sector and the effect of regulation on adoption. This section focuses on exploiting the Census data, while section 5 contains more design for exploiting available public information from the EIA.

4.1 Adoption elasticity estimation

The derived decision rule 2 for technology adoption in Section 2 helps to pin down the key variables of interest: electricity prices, fuel prices, expected change in fuel quantity purchased, expected revenue generated, fixed cost of adoption, and the discount rate. Suppose fuel prices today perfectly forecast fuel prices tomorrow or the uncertainty in future prices can be hedged away in a complete market, but realized shocks persists indefinitely. Suppose further that within any plant, the expected saving in fuel quantity purchased/generated and fixed cost of technology adoption are constant over time. The shock in fuel/electricity prices, therefore, drives the adoption decision. A plant would then adopt the technology at earliest moment when the present value of benefits from adoption exceed that of costs. The option value of waiting is zero. While this assumption is a strong one and perhaps unreasonable for general technology adoption, it is less

so for cogeneration technology since the technology is very mature and existed for hundreds of years. The real fixed cost of installation declined slowly and only started to decline sharply in recent years.

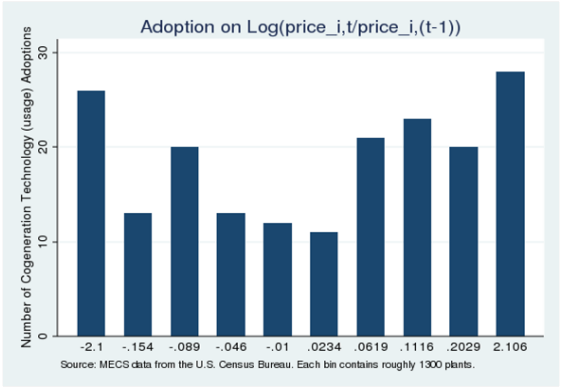
To non-parametrically find evidence (or the lack thereof) on how price shocks impact adoption decisions, I first group plants that do not have cogeneration technology at time t by $\log(\frac{P_{i,t}}{P_{i,t-1}})$, where $P_{i,t}$ is the purchase electricity price plant i faces. Then count the number of plants within each group that adopted the technology between t to $t+1$. Graph the adoption rate on the raw log price ratio. The implicit assumption here is that any plant regardless of age, industry, size, electricity demand (proxy control for expected fuel savings) , survey year, plant hours (proxy control for steam load), when facing the same proportion shock in purchase electricity price, adopt the technology with same probability. To be free of these effects, I also graphed predicted log price ratio on adoption rate. Specially, regress log price ratio on quintile indicators of age, size (measured by total value of shipments), plant hours, electricity usage, and 4-digit NAICS by year fixed effects. This allows for different adoption slopes for each of the observable characteristics. Use the predicted log price ratio tells us similar stories as the raw log price ratio. The left panel of figure 4 shows the adoption rate on raw log price ratio, while the right panel shows the adoption rate on predicted log price ratio. Overall, only a weak link exists between large log price differences and high adoption rates. In this set of figures, it appears that narrow definition of adoption has the strongest positive correlation between adoption rate and price shocks. Figure 5 shows the same idea with log price ratio weighted by energy cost as a share of total material costs ($\frac{P_{i,t}}{P_{i,t-1}} \frac{E_{i,t}+E_{i,t-1}}{2}$) . Notable difference is largely in the broad adoption definition, where there exists a clearer and more significantly positive correlation between price shocks and adoption rate than that of unweighted figures. Weighting by energy cost allows for the fact that more energy intensive plants might be more price sensitive than the less energy intensive guys in making adoption decisions. One can also weight price shocks by electricity cost as a share of total energy costs, which allows for more electricity intensive plants to be more price sensitive. Here, in figure 6, we still observe a stronger pattern between price shocks and adoption rate than that of the unweighted figures, but weaker than that of the figures weighted by energy costs. It could be the case that electricity intensive plants are not energy intensive. This would potentially explain the weaker correlation since the primary objective for plants to adopt the cogeneration technology is not to generate electricity, but to produce electricity as a side product of reusing the manufacturing energy waste. Suppose energy intensity is highly correlated with steam load requirement, the more energy intensive a plant is, the higher the steam load requirement it has, and the more waste heat it generates and can be used in producing electricity by cogeneration. Therefore, the energy intensity might be more relevant for the adoption of cogeneration and not electricity intensity. Overall, these figures show that the relationship between adoption rate and price shocks are unclear, weak (strong in just a few cases), and sensitive to both the definition of technology adoption and the relative weights used for

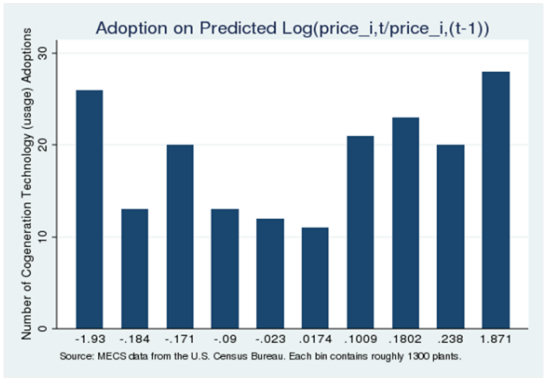
price shocks.

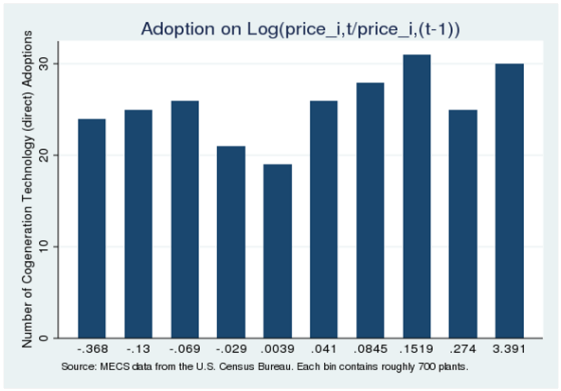
For parametric analysis, I start with a Cox hazard model, then use a cross section linear probability model. The two models give different answers for adoption price elasticities by design. Cox hazard model aims to answer the question from the angle of not-yet-adopters and measures the likelihood of adoption with respect to price shocks. Linear probability model measures sort of the equilibrium cumulative likelihood of technology presence with respect to price levels. The interpretation for the linear probability model is tricky because the sample is not representative of the whole manufacturing industry, with focus on some industries more than others. With a few hundreds of plants sampled with certainty and others (somewhat) at random, more than a third of the plants in total 77,000 plant-year observations only appeared once in the data. The sampling could be correlated with cogeneration technology usage and would bias the estimates, but the direction is unclear. If the certainty plants over-represents the cogeneration plants, then we likely overestimated the adoption elasticity. Likewise, if the certainty plants under-represent the cogeneration plants, we likely underestimated the adoption elasticity in the cross section. If the sample is truly at random so that cogeneration plants and non-cogneration plats are represented fairly, then the interpretation of the estimates would be that how does the average level of electricity prices affect the average likelihood of any plant in the manufacturing sector having cogeneration technology present. This potential sampling bias is of concern, particularly when comparing to both EIA and ICF data. While EIA and ICF both show an increase in adoption in the manufacturing sector between 1980 to 2005, and a decline after, the Census data show roughly consistent number of plants having cogeneration technology. Out of roughly 4000 plant-year observations reporting having technology on site, the number of plants reporting with the technology first decreased from 1994 level, then increased only slightly each year after 1988 (See table 2). Due to confidentiality reasons, the numbers are rounded to the nearest hundred. If we were to expect ICF and EIA data being correct, we would expect to see the number of cogeneration plants increase till 2005 and become steady afterwords. But the Census data seems to suggest the peak was in 1994 and many reversed the adoption. Therefore, it is very likely that cogeneration plants are under represented in the Census sample. We would expect the estimates from linear probability model be biased downwards and less than the estimates obtain in the Cox hazard model.

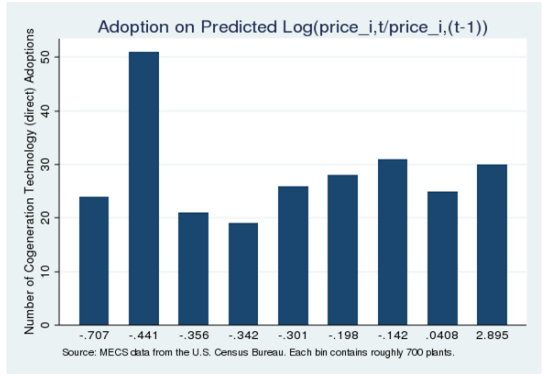
In the hazard model analysis, I treat plants that have not adopted the cogeneration technology by 2010 as censored. Once a plant becomes an adopter, it is no longer in the sample for the potential adoption in the later periods as "at-risk" in the Cox model. The adoption event is modeled as an absorbing one, which happens to a plant at most once. This is not far from what is observed in the data. Only a very small number of plants adopted more than once by usage adoption definition, even smaller number of plants by direct adoption definition. The first adoption time is defined to be the time the plant enters into the absorbing

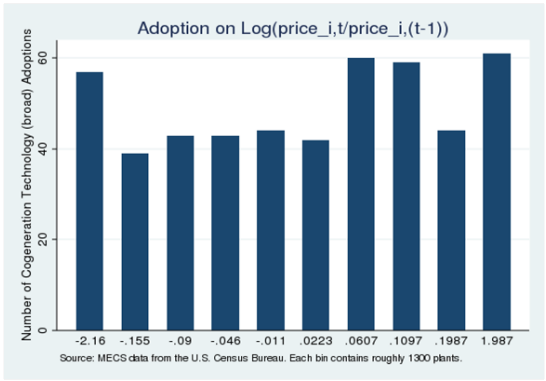
Figure 4: Adoption Rate on Price Shocks

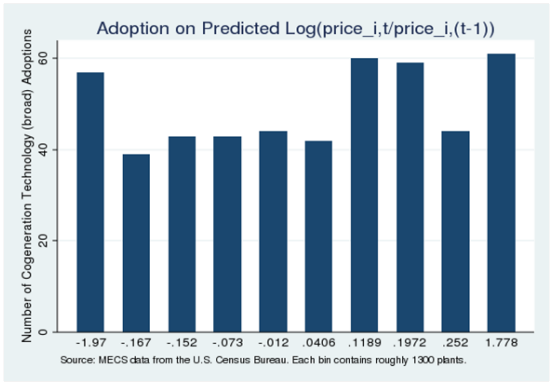


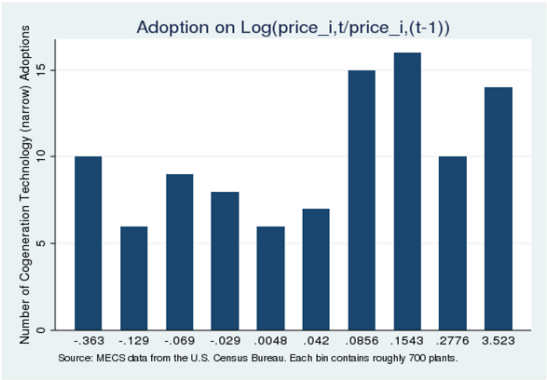


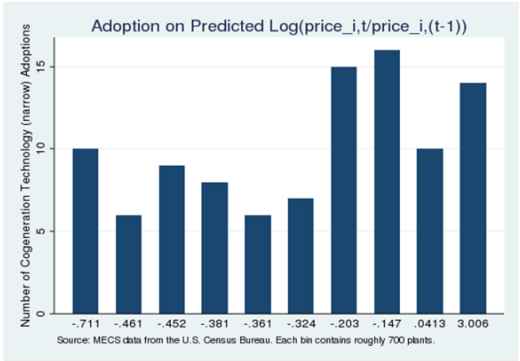






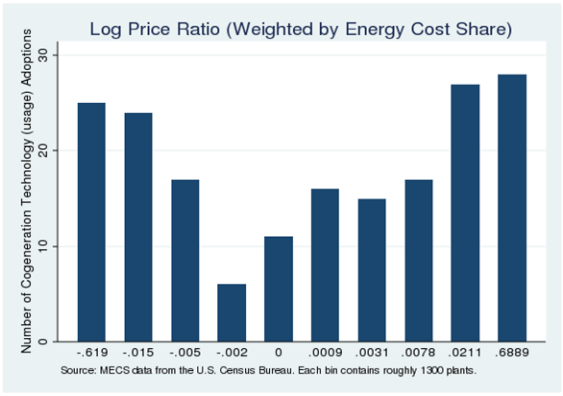


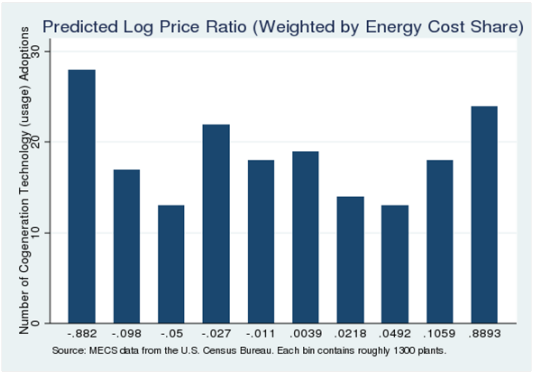


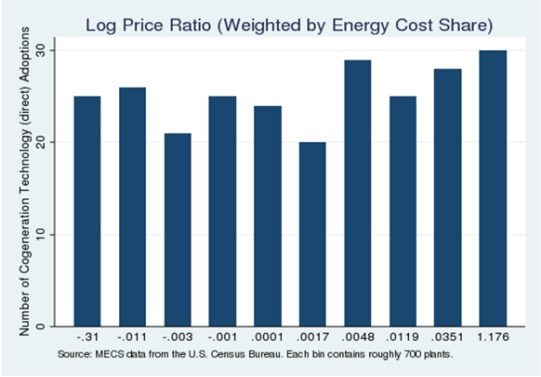


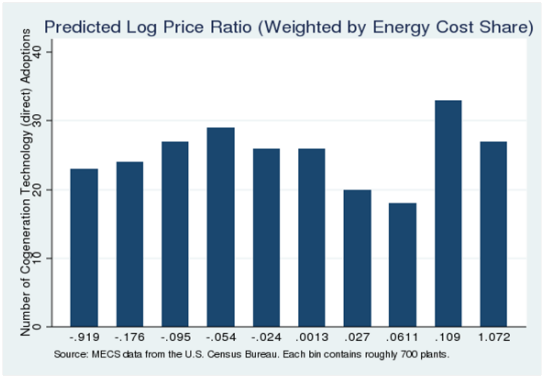
Notes: Data is from the US Census Bureau. Each number on x-axis represents the mean log price difference in the perspective bin. The left panel is the adoption rate on raw data, while the right panel is the adoption rate on predicted values. Data covers from 1988 to 2006 for usage adoption and from 1994 to 2006 for direct adoption.

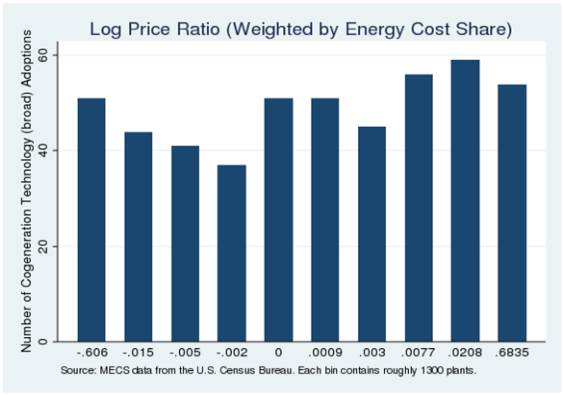
Figure 5: Adoption Rate on Price Shocks Weighted by Energy Cost Share

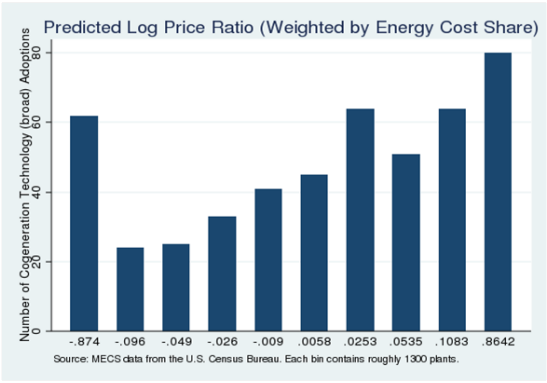


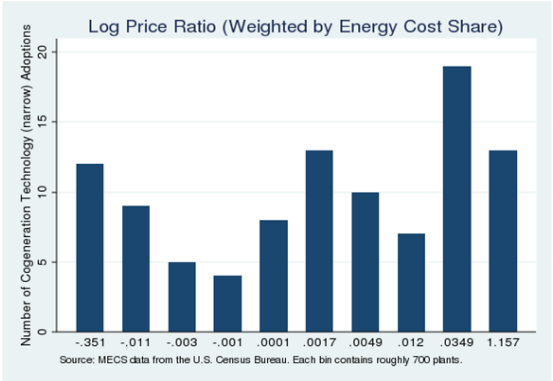


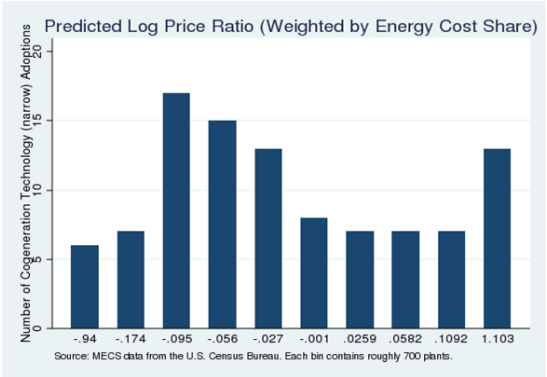






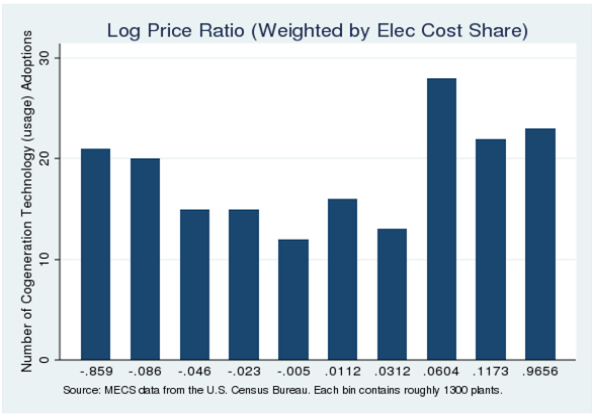


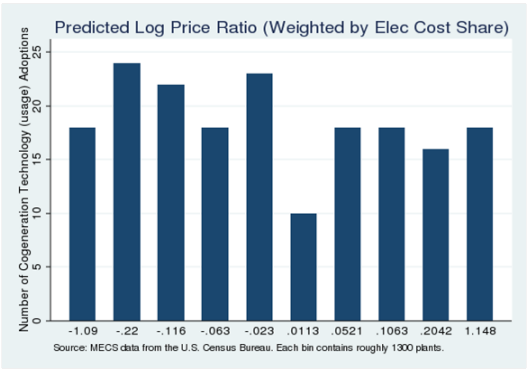


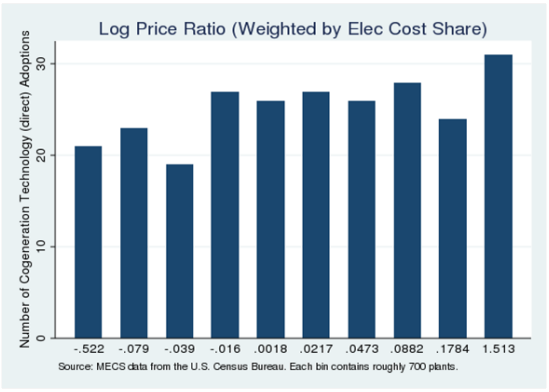


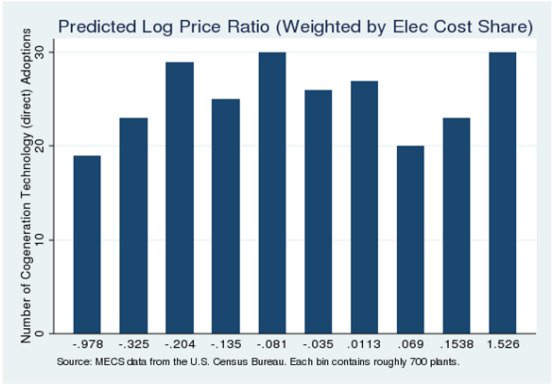
Notes: Data is from the US Census Bureau. Each number on x-axis represents the mean log price difference in the perspective bin. The left panel is the adoption rate on price data weighted by energy cost share (energy cost as a share of total material cost), while the right panel is the adoption rate on predicted values. Data covers from 1988 to 2006 for usage adoption and from 1994 to 2006 for direct adoption.

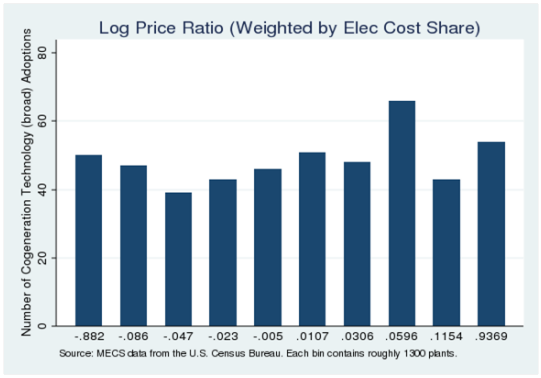
Figure 6: Adoption Rate on Price Shocks Weighted by Electricity Cost Share

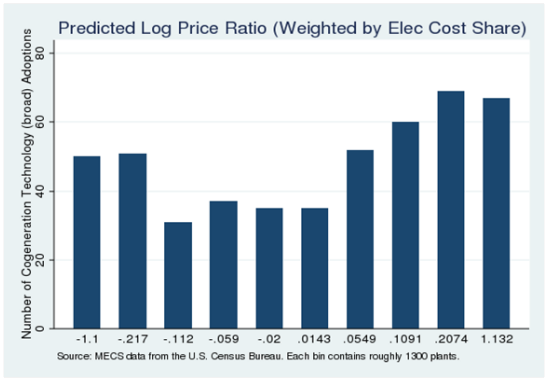


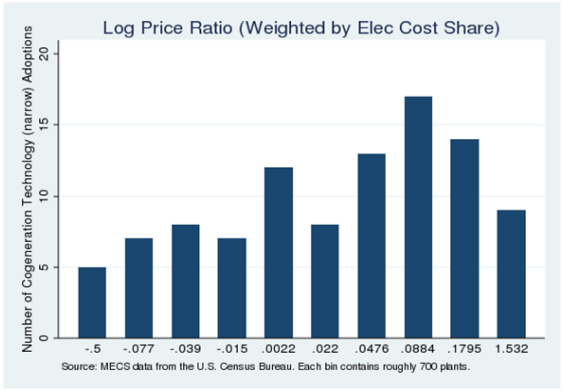


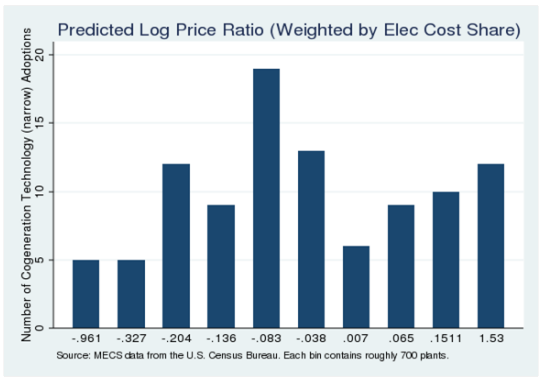












Notes: Data is from the US Census Bureau. Each number on x-axis represents the mean log price difference in the perspective bin. The left panel is the adoption rate on price data weighted by electricity cost share (electricity cost as a share of total energy cost), while the right panel is the adoption rate on predicted values. Data covers from 1988 to 2006 for usage adoption and from 1994 to 2006 for direct adoption.

Table 2: Rough Number of Cogeneration Plants and Concentration Ratio in MECS

	Penetration Ratio			Num of Est.			
	(1) TVS	(2) TE	(3) Num	(4) Direct	(5) Usage	(6) Broad	(7) Narrow
1985	0.044	0.101	0.043	–	500	500	–
1988	0.059	0.130	0.056	–	600	600	–
1991	0.056	0.130	0.052	–	600	600	–
1994	0.082	0.157	0.078	900	600	1000	500
1998	0.103	0.180	0.099	600	500	700	400
2002	0.082	0.179	0.077	600	400	700	400
2006	0.122	0.208	0.116	600	400	600	400
2010	0.111	0.201	0.106	600	400	600	400

Notes: Data is from the US Census Bureau. First three columns are penetration ratios. Column 1 is measured by total value of shipment. Divide the total value of shipments from the plants with cogeneration technology (technology usage definition: cogenerated nonzero electricity on site) by total value of shipments from all the plants in a given year. Similarly, column 2 is measured by total employment. Column 3 is measured by number of establishments. Last four columns are number of establishments with cogeneration technology in a given year. Column 4 uses direct technology definition: plant reports to have cogeneration technology on site. Column 5 uses usage technology definition: plant cogenerates nonzero electricity on site. Column 6 uses broad definition, which is the union of the direct and usage definition. Column 7 uses narrow definition, which is the intersection of the direct and usage definition. Due to confidentiality protection, the numbers are rounded to the nearest hundred.

state. Plants that have missing observations in some survey years between 1985 and 2010 are considered as truncation records. Their contribution to the overall likelihood are properly dealt with. The following semiparametric Cox proportional hazard model (Cox, 1972) is used as baseline model in the analysis:

$$\lambda_{it} \quad = \quad \lambda_0(t) \exp[\beta \Delta P_{ijt} + \mathbf{1}_j + X_{it}] \tag{3}$$

where λ_{it} is the hazard rate for plant i, the conditional probability that plant i adopts the cogeneration technology at time t, given that it has not yet adopted the technology. $\lambda_0(t)$ is a fully flexible, nonparametric baseline hazard and is specified with respect to calendar time as well. Hence, no year fixed effect is included in the model. $\Delta P_{ijt} = P_{ijt} - P_{ijt-1}$ is the log price difference of purchased electricity plant i faces from time t to $t-1$. ΔP_{ijt} is assumed to be exogenous to the plants. $\mathbf{1}_j$ is a fixed effect on four digit NAICS industry classification. X_{it} consists a rich set of plant level controls at time t, including plant hours, age, size (value of shipments), and electricity usage (all as quintile indicator, so no linearity is assumed). Electricity usage captures the electricity demand and plant hours somewhat captures the steam load. Both attempt to control for the expected savings in quantity of purchased electricity.

β here is our parameter of interest. It is identified through spacial variation of otherwise similar plants facing different exogenous price shocks. Most of the spacial variation is coming out of location: plants face different price shocks in different states. Suppose two plants A and B are identical in all the control variables, and in the same industry, A faces a price increase that is 1 unit higher than B. $\frac{\lambda_{At}}{\lambda_{Bt}} = \frac{\exp(\beta(\Delta P_{ijt}+1))}{\exp(\beta \Delta P_{ijt})} = \exp(\beta)$. 1 unit here is equivalent to e ($\approx$2.718) times the price difference relative to B. In other words, if price today

is 4 times the price yesterday for B ($\log(\frac{4}{1})$), for A to have a hazard rate of $\exp(\beta)$ relative to B, A has to face a price increase of $4 \times 2.718 \approx 10$ times ($\log(\frac{4}{1} \times e) = \log(\frac{4}{1}) + 1$) of that yesterday. If $\exp(\beta) < 1$, A faces a hazard rate that is only $\exp(\beta) \times 100$ percent of the hazard faced by B; B has a higher hazard rate. If $\exp(\beta) > 1$, A is then $(1 - \exp(\beta)) \times 100$ percent more likely to adopt the technology at time t.

Table 3 reports the estimates for $\exp(\beta)$ (Hazard Rate) in various specifications. The additional specifications are to control for different fuel usage and prices the plants face, since instead of adopting an energy-preserving technology, the plant can always choose to switch fuels. β is consistent across within each adoption definition as a robustness check. However, across adoption definitions, β values vary and becomes statistically insignificant at times. Narrow adoption definition shows the largest response of adoption to price shocks. On average, the log electricity price difference is close to 0.36, which means the price of today is about 1.43 times the price of yesterday. Using the results from narrow adoption definition, one standard deviation (1.45) increase in log price difference above the mean (0.36) would translate into a 10 percent ($\exp(\beta \times (1.45 + 0.36))/\exp(\beta \times 0.36) \approx \exp(0.0677 \times 1.81)/\exp(0.0677 \times 0.36)$) increase in probability of adoption. One can also interpret the results as that for any plant faces a price increase of four fold ($1.43 \times e$), it is 7 percent more likely than the average plant to adopt the cogeneration technology. The mean price level in the sample is roughly 6 cents per kwh for purchased electricity. Increase that by four fold would give us a price level of 24 cents per kwh. This is quite high. So even though 7 percent is large and significant (since in the overall cross section, only slightly more than 7 percent of the plants reported having the cogeneration technology on site), the price incentive would need to be at a somewhat implausible range. Alternatively, we can also translate the results to that for a plant to increase the likelihood of adoption by 1 percent more than the average adoption probability, the electricity price today has to be 1.158 times more than that of yesterday, going from 6 cents to 7 cents. Usage, direct, and broad adoption, however, seems to suggest ambiguous results in both the magnitude and direction. Not much can be inferred from these numbers. Since the ultimate goal for adoption is to use the technology, narrow definition captures both. So it is our preferred estimate. Somewhat puzzling is that energy cost share does not seem to have the impact on adoption decisions as much as it suggests from the previous nonparametric analysis figures. There seems to be no reason to believe energy cost shares interact with price shocks nonlinearly. Perhaps the cost shares are already captured in the quintile indicators of fuel consumption, or other plant characteristics in the control variables. So we do not see the effect we expected.

The baseline specification for cross section linear probability model is

$$A_{ijt} = \alpha + \beta P_{ijt} + \mathbf{1}_j \times \mathbf{1}_t + X_{it} \tag{4}$$

Table 3: Cox Semiparametric Proportional Hazard Model Regression

	Narrow Adoption				Usage Adoption	Direct Adoption	Broad Adoption
	(1)	(2)	(3)	(4)			
Log elec. price difference ΔP_{ijt}	1.0715**	1.0717**	1.0688**	1.0688**	0.9991	1.0307	0.9737~
	(0.0234)	(0.0236)	(0.0233)	(0.0231)	(0.0452)	(0.0231)	(0.015)
Energy cost share E_{ijt}			0.9990	0.9988	1.0019	1.0074**	0.9956
			(0.0158)	(0.0151)	(0.0040)	(0.0021)	(0.0080)
Weighted price differences $\Delta E_{ijt} P_{ijt}$			1.0304	1.0349	1.0194	0.9959	1.0327
			(0.0437)	(0.0434)	(0.0338)	(0.0239)	(0.0218)
Quintile indicator for X_{it}	X	X	X	X	X	X	X
Indicator on fuel type		X		X	X	X	X
Interaction (fuel type X fuel price)		X		X	X	X	X
No. of obs (approx.)	12000	12000	12000	12000	20000	12000	20000
No. of subjects (approx.)	6500	6500	6500	6500	8700	6500	8700
No. of failures (approx.)	200	200	200	200	300	400	700

Notes: Hazard Ratio is reported instead of the coefficients. Robust standard errors clustered at the state level are in parentheses. Data is from U.S. Census Bureau. All figures are unweighted. Table reports hazard model regressions, with failure is defined as adoption. Treat plants that have still not adopted the cogeneration technology by 2010 (the end of our sample period) as censored. All regressions include 4-digit industry fixed effect. X_{it} consists a rich set of plant level controls at time t, including plant hours, age, size (value of shipments), and electricity usage. We do not include year fixed effect because in the Cox model, the fully flexible baseline hazard is also specified with respect to calendar time. Direct technology adoption equals 1 if a plant reports to have cogeneration technology on site but not in the previous survey year. Usage technology adoption equals 1 if a plant cogenerates nonzero electricity on site but cogenerated zero electricity in the previous survey year. Broad definition is the union of the direct and usage definition. Narrow definition is the intersection of the direct and usage definition. Due to confidentiality protection, number of observations are rounded to the nearest thousand. Each observation is a manufacturing plant in the survey. ~ p <0.1, * p<0.05, ** p<0.01

where A_{ijt} is an indicator for cogeneration technology presence at plant i in industry j at time t. P_{ijt} is the log of purchased electricity price plant i faces at time t. X_{it} are defined the same way as in the hazard model. $\mathbf{1}_t \times \mathbf{1}_j$ is 4-digit NAICS by year fixed effect. Parameter of interest β is again identified through spacial variation. Since the dependent variable here is the presence of technology, not the adoption of technology, it makes sense to switch the price difference to price levels, because price differences are not indicative of a state of being, but a change in the state. The likelihood of having the technology presence is expected to be increasing the level of purchased electricity price. This is indeed what we observe in the regressions and consistent across various adoption definitions. Table 4 shows the regression results. Direct technology and narrow technology definitions give almost identical estimates, while usage technology and broad technology definition gives similar estimates. The direction for which energy prices moves the technology presence is within our expectation. However, the magnitude is puzzling, particularly when comparing across definitions. If we were to increase the price of electricity by 1 cent, from 6 cent kwh to 7 cent kwh, direct technology predicts an increase in the likelihood of a plant having the technology present by 0.7 percent ($0.046 \times \log(\frac{7}{6})$). Usage/broad technology definition, on the other hand, with the same price increases, predicts only an increase of 0.06 ($0.004 \times \log(\frac{7}{6})$) percent in the likelihood of technology presence. This difference in relative magnitude can be arriving from two aspects. First is from the definition. Usage definition encompasses two dimensions of the technology, the presence and utilization. Conditional on technology presence, the utilization should be relatively unaffected by electricity price shocks, since the utilization is based on steam load at the plant as a feature of the technology. The direct definition only considers the presence of technology, so it is more responsive to the price shocks. Second is from the sample differences. Usage definition sample has more than 50 percent more observations than direct definition sample. Per discussion at the beginning of this section, samples are not drawn randomly. It could be the case that the more observations we have, the more bias we have. But this argument is not very convincing because the extra sample points in usage definition are largely from non-certainty plants (which are drawn more randomly than certainty plants). Therefore, because the main objective here is to study the adoption elasticities, not necessarily the utilization elasticities, our preferred estimate for the linear probability analysis is also using narrow technology definition as in the Cox hazard model analysis. For a 1 cent increase in electricity purchase price from 6 cents to 7 cents per kwh, linear probability model suggests an increase of technology presence by 0.7 percent and Cox model suggests 1 percent.

Another specification uses the same right hand side variables, but change the dependent variable to the raw count of the number of cogeneration technologies on site. The MECS survey starting from 1994 asks for the presence of five different types of cogeneration technologies. Previously, $A_{ijt} = 1$ if any of these five is present on site. By looking at the number of cogeneration technologies, we treat all these technologies

Table 4: Cross Section Linear Probability Regression

Dep: A_{ijt}	Direct Technology Presence			Usage Technology Presence			Narrow Technology Presence		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Log(price of elec)	0.0428**	0.0464**	0.0465**	0.0039**	0.0038**	0.0039**	0.0457**	0.0486**	0.0487**
	(0.0081)	(0.0089)	(0.0089)	(0.0007)	(0.0007)	(0.0007)	(0.0080)	(0.0087)	(0.0086)
Log(price of natural gas)			-0.0031			-0.0039~			-0.0021~
			(0.0023)			(0.0022)			(0.0018)
Quintile indicator for X_{it}	X	X	X	X	X	X	X	X	X
Indicator on fuel type		X	X		X	X		X	X
Interaction (fuel type X fuel price)			X			X			X
N (approx.)	43,000	43,000	43,000	76,000	76,000	76,000	43,000	43,000	43,000
R^2	0.2618	0.2884	0.2889	0.2617	0.2846	0.2856	0.2582	0.2893	0.2904

	Broad Technology Presence			Number of Cogen Tech		
	(1)	(2)	(3)	(1)	(2)	(3)
Log(price of elec)	0.0038**	0.0037**	0.0037**	0.0696**	0.0738**	0.0740**
	(0.0007)	(0.0007)	(0.0007)	(0.0127)	(0.0133)	(0.0132)
Log(price of natural gas)			-0.0037			-0.0060
			(0.0026)			(0.0046)
Quintile indicator for X_{it}	X	X	X	X	X	X
Indicator on fuel type		X	X		X	X
Interaction (fuel type X fuel price)			X			X
N (approx.)	76,000	76,000	76,000	43,000	43,000	43,000
R^2	0.2709	0.2913	0.2920	0.2218	0.2313	0.2318

Notes: Standard errors clustered at the state level are in parentheses. Data is from U.S. Census Bureau. All figures are unweighted. Table reports regressions of the pooled cross section (from years 1985-2010) indicator of cogeneration technology on corresponding plant level controls. Column $1X_{it}$ consists a rich set of plant level controls at time t, including plant hours, age, size (value of shipments), and electricity usage. All regressions include a constant and year by 4-digit NAICS fixed effect. Direct technology adoption equals 1 if a plant reports to have cogeneration technology on site but not in the previous survey year. Usage technology adoption equals 1 if a plant cogenerates nonzero electricity on site but cogenerated zero electricity in the previous survey year. Broad definition is the union of the direct and usage definition. Narrow definition is the intersection of the direct and usage definition. Due to confidentiality protection, number of observations are rounded to the nearest thousand.Each observation is a manufacturing plant in the survey. ~ p <0.1, * p<0.05, ** p<0.01

symmetrically and assume that the technologies were perfect substitutes. The estimates give us the same magnitude as the Cox model: with a 1 cent increase in price from 6 cents to 7 cents, the likelihood of having one more cogeneration technology present on site increases by about 1 percent. This is comforting as it not only confirms the robustness of the elasticity parameters we estimated using different models, but also confirms the understanding that cogeneration technologies are likely close substitutes of one another.

4.2 Effects of adoption

For potentially providing a recommendation on the socially optimal level of cogeneration adoption, understanding the effects of adoption is almost as important as understanding what drives adoption. Adoption at individual non-utility plant has effects on both the plant itself and the utility sector. It leads to a change in total fuel consumption and carbon emissions for electricity generation purposes and impacts the average electricity prices. Some of these effects are internalized by the non-utility plants when making adoption

decisions, while others are not, particularly for carbon emissions. The desired layout for this subsection is to split evenly on the internalized effects and externality effects. Currently, I can only report on how manufacturing plants changed their fuel consumption pattern prior and after the adoption of cogeneration. It is part of the research agenda to complete the picture in external adoption effects.

We use a simple event study empirical design to study the effects of adoption on manufacturing plants:

$$\Delta y_{ijt} = \sum_{a=-3}^{3} \alpha_a A_{ij(t+a)} + X_{it} + \mathbf{1}_j \times \mathbf{1}_t$$

where Δy_{ijt} is the log first difference of the variable of interest, here we use the log values of TFP, total value of shipment, total employment, total electricity usage, quantity of purchased electricity, and quantity of purchased natural gas (since 70 percent of cogeneration technology use natural gas as primary fuel input). X_{it} and $\mathbf{1}_j \times \mathbf{1}_t$ are defined the same way as in the previous regressions.[7] Event is defined as the adoption or dis-adoption of a cogeneration technology. Event window covers three pre-event periods and three post-event periods. Each event period is roughly 3-4 years corresponding to survey frequencies. Since there are only 8 MECS samples, to be in the regressions, a plant has to have a longitudinal dimension that covers at least 7 of the survey years[8]. This reduces our sample to roughly 600 plants. Again, if we were to merge the EIA data with the Census data, we would have a much larger longitudinal panel. But with the small sample we do have, results demonstrated significant change of fuel consumption patterns for adopting plants.

Not surprisingly, adoption of cogeneration technology has small effects on the shipment, or employment, and no effect on the TFP (see figure 8). The primary purpose of the technology is not to alter the main business model of the manufacturing plants, but to use the waste in the manufacturing process to defray electricity costs and potentially generate some revenue by selling excess electricity. There is also not much effect on the amount of electricity used on site. This again confirmed our understanding. The large effects are seen in quantity of purchased electricity and quantity of purchased natural gas (see figure 7). Both effects are expected since the adopting plants no longer need to purchase as much electricity from the utilities, but now needs to purchase more natural gas since more fuel is required to generate the same amount of useful thermal output as before.

Figure 7 and 8 shows the results graphically and table 5 reports the coefficients on selected variables. Immediately going from not having the technology to adoption (column (2) in table 5), a plant experienced a decrease in quantity of purchased electricity by 75 percent (1-exp(-1.3556)) at event time 0. The effect

[7] Plant controls suppose to be in differences as well in the specification. However, X_{it} are quintile indicators and are not continuous. We are allowing for differential trends in adoption effects for each quintile.

[8] For this reason, here we use usage definition for A_{ijt} since direct/narrow definition do not have enough survey year coverage. Using usage definition is reasonable because we are studying the effect of fuel consumption change due to the utilization of the cogeneration technology.

persists through all three post periods. The cumulative effects amounts to a total reduction of 90 percent from pre-adoption period $(1 - \exp(-1.3556)\exp(-0.6796)\exp(-.2202)\exp(-.0978))$. Similarly, the cumulative effects of adoption amount to an increase in the quantity of purchased natural gas by 6 times than that of the level in pre-adoption period $(\exp(1.5629)\exp(0.3644)\exp(-0.0321)\exp(-0.0608))$. The effect immediately after adoption at event time 0 is close to a 5-time $(\exp(1.5629))$ increase in fuel consumption. If we were to include reserve-adoption events in the sample (column (1) in table 5), the effects are not as large, but still significant. The adopting plants would decrease purchased electricity quantity by close to 50 percent, increase purchased natural gas by 68 percent right after adoption, and sustain the change through later periods. These numbers are a little difficult to interpret since we do not know if the sharp increase in natural gas consumption is also due to fuel switching. For example, one can imagine a plant prior adoption uses natural gas, but not as primary fuel, which supposes to be coal, for manufacturing production. After adoption, the plant decides to switch primary fuel to natural gas. This would lead to a sharp increase in natural gas consumption, and should not be interpreted as a pure substitution effect between purchased electricity and purchased natural gas. If the change is driven entirely by the proposed substitution effects, we can then compute the efficiency cogeneration technology provides in converting natural gas to electricity. This efficiency can be used to compare with that of the utility power plants to help with determining the socially optimal level of cogeneration adoption. We suspect the increase in natural gas consumption is largely driven by fuel switching and less so by the substitution effect. More analysis needs to be done to confirm our hypothesis.

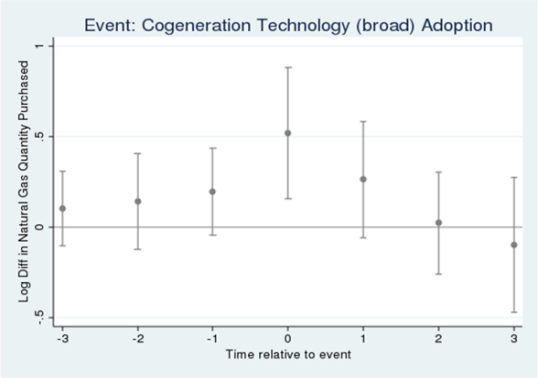
Nonetheless, due to the large decrease in purchased electricity, the demand for electricity will go down for areas with many cogeneration plants. Since the electricity is mostly a side product, cogeneration plants can sell the excess electricity at lower price than central power plants. Both demand and supply factors should drive down the cost of electricity locally. Very preliminary results do show a reduction in electricity prices in 3-digit zip code areas with more cogenerated electricity, using the within zip code variation across time. [The results are not reported because it is a recent finding and need to receive disclosure clearance.]

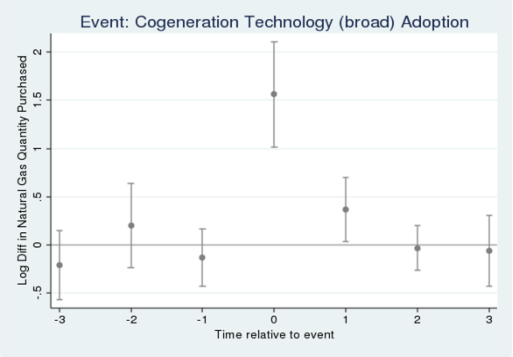
The effects on utilities and carbon emissions are not yet studied but are on our schedule. The effects we studied so far are significant for changing fuel consumption behavior of the adopting plants in the manufacturing sector and are expected to have important externalities to be considered.

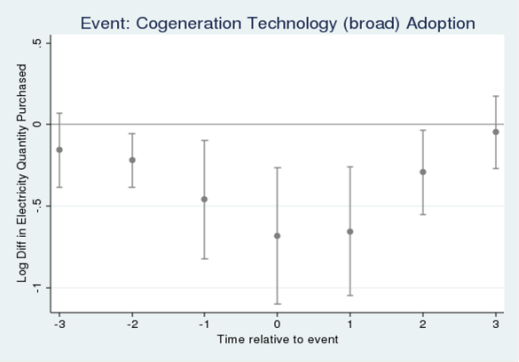
5 Conclusion and Future Plan

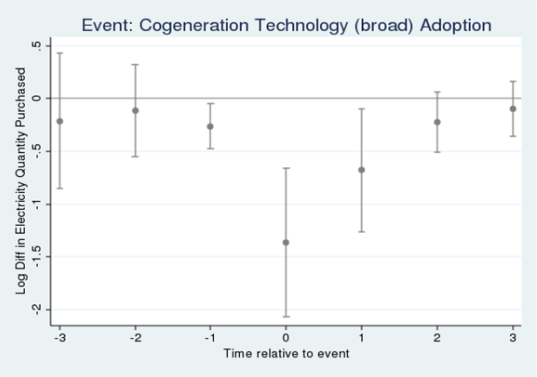
Preliminary results show that electricity and fuel prices might not be a deciding factor for cogeneration technology adoption in the US manufacturing sector. With a 1 cent increase in electricity prices, going from

Figure 7: Effect of Adoption on Manufacturing Plants



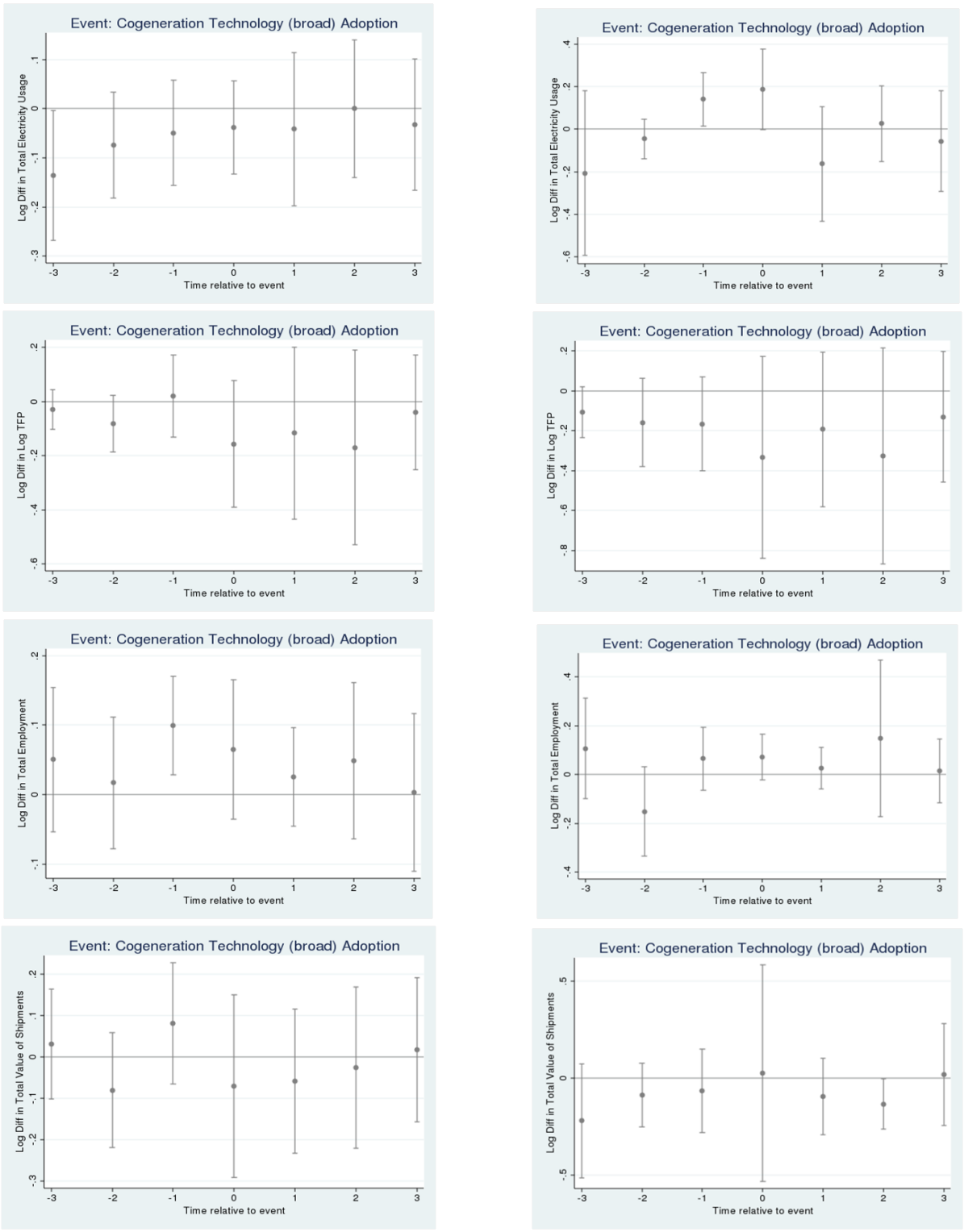






Notes: Data is from the US Census Bureau. Event time is zero when a plant reports to have cogenerated nonzero amount of electricity on site for the first time. The left panel defines event equals 1 if a plant goes from being a non user to a user of the technology, -1 if goes from user to non user; while the right panel does not include the -1 events in the left panel. Data covers from 1988 to 2010. N is roughly 600 for the left panel and 500 for the right panel.

Figure 8: Effect of Adoption on Manufacturing Plants



Notes: Data is from the US Census Bureau. Event time is zero when a plant reports to have cogenerated nonzero amount of electricity on site for the first time. The left panel defines event equals 1 if a plant goes from being a non user to a user of the technology, -1 if goes from user to non user; while the right panel does not include the -1 events in the left panel. Data covers from 1988 to 2010. N is roughly 600 for the left panel and 500 for the right panel.

Table 5: Effect of Adoption on Manufacturing Plants

Dep:	Δquantity of purchased electricity		Δ quantity of purchased natural gas	
	(1)	(2)	(1)	(2)
Event time 0	-0.6822**	-1.3556**	0.5182**	1.5629**
	(0.2082)	(0.3493)	(0.1804)	(0.2709)
[Cumulative effect]	[0.5055]	[0.2578]	[1.6790]	[4.7726]
Post-event time 1	-0.6547**	-0.6796*	0.2630~	0.3644*
	(0.1960)	(0.2880)	(0.1600)	(0.1651)
[Cumulative effect]	[0.2627]	[0.1307]	[2.1841]	[6.8709]
Post-event time 2	-0.2921*	-0.2209	0.0230	-0.0321
	(0.1291)	(0.1414)	(0.1402)	(0.1161)
[Cumulative effect]	[0.1961]	[0.1048]	[2.2349]	[6.6539]
Post-event time 3	-0.0463	-0.0978	-0.0964	-0.0608
	(0.1110)	(0.1294)	(0.1849)	(0.1832)
[Cumulative effect]	[0.1872]	[0.0951]	[2.0295]	[6.2614]
Quintile indicator for X_{it}	X	X	X	X
Indicator on fuel type	X	X	X	X
Interaction (fuel type X fuel price)	X	X	X	X
N (Approx.)	600	500	600	400
R^2	0.5286	0.6051	0.2551	0.3852

Notes: Standard errors clustered at the state level are in parentheses. Data is from U.S. Census Bureau. The column (1) defines event equals 1 if a plant goes from being a non user to a user of the technology, -1 if goes from user to non user; while column (2) does not include the -1 events in the column (1). All figures are unweighted.X_{it} consists a rich set of plant level controls at time t, including plant hours, age, size (value of shipments), and electricity usage (all as quintile indicator). All regressions include year by 4-digit NAICS fixed effect. Due to confidentiality protection, number of observations are rounded to the nearest hundred. ~ p <0.1, * p<0.05, ** p<0.01

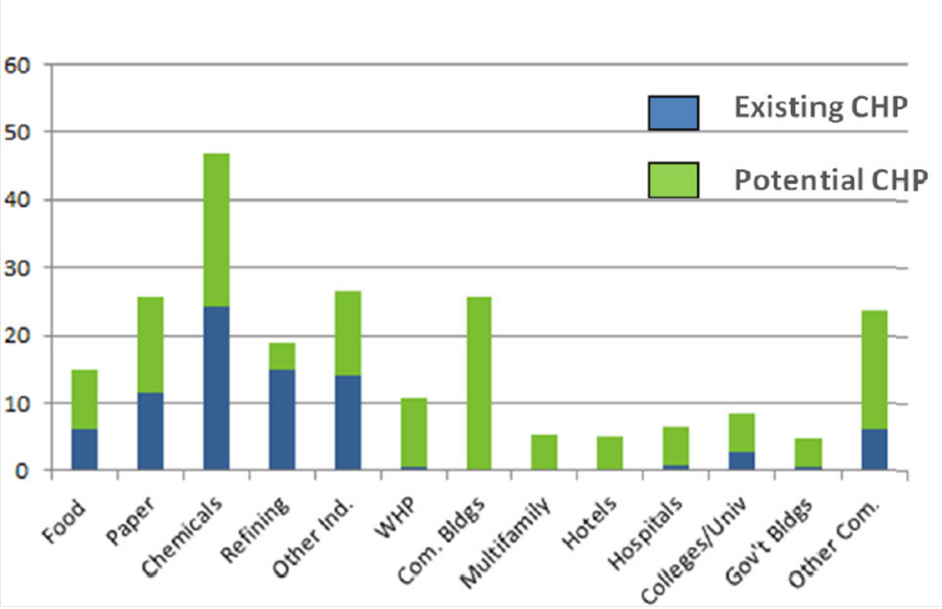
6 cents per kwh to 7 cents per kwh, the likelihood for cogeneration adoption increases by 1 percent. We hope to further study the impact of regulatory factors on the adoption of cogeneration technology, using more data from the EIA. The plants that did experience an adoption event reduce their purchase quantity of electricity immediately by at least 50 percent and increase their purchase quantity of natural gas by at least 68 percent. This change of fuel consumption pattern persists through later periods, while shipment, employment, and TFP remain largely unchanged with the adoption. The industrial average electricity purchase price decreases with an increase in the amount of cogenerated electricity.

There are much more needed to be done. Top on the list is to estimate the effects of cogeneration usage on utility sector as well as on carbon emissions overall. Second is to better understand the role regulatory factors play in the adoption decision as well as the utilization of the technology. I plan to use reduced form methods to study these two proposed questions. Both need to exploit the EIA data. Variation from within 3-digit level zip code across time can be used to address the effect of adoption on local utilities and environmental factors. After 2005, cogeneration plants with more than 20 MW nameplate capacity loses the "must-purchase" privilege. The discontinuity around this 20 MW figure can be used for regression discontinuity design and estimate the effect of this "must-purchase" regulatory obligation on various plant outcomes. It may also help to explain the sharp drop in adoption after 2005.

To complete the analysis, particularly to provide policy recommendations for the optimal cogeneration investment level, some measures on cost are needed. It is not yet clear what the best approach to this is. One way is to simply request the confidential data from the EIA, which has measures on equipment costs. But it might take more than three months to receive clearance. Alternatively, we can try to merge EIA onto Census data on plant level by year. Census has good manufacturing data with capital investment figures readily available while EIA can supplement with better data in electricity generators (better in both frequency and coverage than MECS). The downside is that EIA's data covers not only the manufacturing sector. We are losing information on overall cogeneration employment by merging plants only in the manufacturing sector. While at the current moment, over 70 percent of cogeneration capacity is in manufacturing, more than half of the technical potential capacity for future adoption is sectors outside manufacturing. A complete model that takes into consideration for both investment in utility sector and non-utility sector is still needed, particularly in generating counterfactual simulation.

If we were to complete the desired goals, we would be in a position to be able to explain the observed pattern in cogeneration adoption in the US, as well as addressing the general equilibrium effects of cogeneration adoption on electricity prices and carbon emissions, which both affect consumers in important ways.

Figure 9: Technical Potential for CHP Adoption



Source: U.S. Department of Energy, CHP: A Clean Energy Solution

Appendix

One of the potential reasons behind the drop in adoption rate after 2005 could be because the market at this point has been penetrated by the technology. This is not the case by US Department of Energy estimates, nor by author's calculation from the MECS data from US Census Bureau. Figure 9 from the US Department of Energy shows that we could still increase the amount of cogeneration capacity in the manufacturing sector by a factor of two, and could increase cogeneration capacity in other sectors by even more, with largest potential in commercial buildings that amounts to over twenty. Table 6 shows the top ten 3-digit NAICS industries that cogenerated largest amount of electricity in year 1985 and 2010. The ranking has not changed much at all between the 15 years. In many of these industries where cogeneration technology were exploited the most, the penetration ratio (across different measures) was low in 1985 and remained low in 2010 (except for paper manufacturing, and petroleum and coal products manufacturing). This suggests that even among the most penetrated industries, there is still space for adoption. Outside of top three industries, no industry has a penetration ratio higher than 20 percent by all measures in 2010. Of course, a more careful cost benefit analysis would be needed to estimate the adoption potential. But from the technical side (plants with high steam load factor that could use the waste heat for electricity generation), there is ample room to grow.

Table 6: Penetration Ratio by Top Cogen Industries

1985 Ranking	3-digit NAICS Industry Manufacturing	Penetration Ratio (1)	(2)	(3)	2010 Ranking	3-digit NAICS Industry Manufacturing	Penetration Ratio (1)	(2)	(3)
1	Paper	0.2638	0.4698	0.2585	1	Chemical	0.2663	0.4291	0.2512
2	Chemical	0.0868	0.2544	0.0779	2	Paper	0.5520	0.7544	0.5462
3	Petroleum and Coal Products	0.1204	0.2982	0.1105	3	Petroleum and Coal Products	0.2884	0.7586	0.2254
4	Primary Metal	0.0424	0.1908	0.0391	4	Primary Metal	0.0743	0.1950	0.0672
5	Food	0.0629	0.1026	0.0645	5	Food	0.1229	0.1315	0.1224
6	Beverage and Tobacco Product	0.0759	0.1968	0.0735	6	Wood Product	0.1014	0.1515	0.10
7	Transportation Equipment	0.0116	0.0190	0.0113	7	Transportation Equipment	0.1219	0.1995	0.1119
8	Wood Product	0.0618	0.0994	0.0627	8	Beverage and Tobacco Product	0.1346	0.2369	0.1229
9	Textile Mills	0.0282	0.1063	0.0272	9	Nonmetallic Mineral Product	0.0454	0.0548	0.0430
10	Nonmetallic Mineral Product	0.0074	0.0271	0.0071	10	Plastics and Rubber Products	0.0907	0.1785	0.0917

Notes: Data is from the US Census Bureau. Ranking is the top ten 3-digit NAICS industries with largest amount of cogenerated electricity, 1 being the highest. Under column Penetration Ratio: Column 1 is measured by total value of shipment. Divide the total value of shipments from the plants with cogeneration technology (technology usage definition: cogenerated nonzero electricity on site) by total value of shipments from all the plants in a given year. Similarly, column 2 is measured by total employment. Column 3 is measured by number of establishments.

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